BISOPROLOL

HFrEF mortality benefit

Do not stop suddenly

CHEMICAL NAME (RS)-1-{4-[(2-isopropoxyethoxy)methyl]phenoxy}-3-(isopropylamino)propan-2-ol (fumarate)

CLASS Highly cardioselective β1-adrenoceptor blocker

TRADE NAMES Concor · Cardicor · Zebeta · Emcor

THERAPEUTIC USES / INDICATIONS

- Heart failure with reduced EF (stable) — ↓ mortality.
- Hypertension.
- Stable angina.
- Atrial fibrillation rate control.

ADMINISTRATION

- **PO** once daily in the morning, with or without food.
- HF: start low and titrate slowly every 2 weeks.

DOSAGES

- **Hypertension/angina:** 5–10 mg once daily (max 20 mg).
- **Heart failure:** 1.25 mg once daily, double every 1–2 weeks as tolerated to target 10 mg.
- **AF:** 2.5–10 mg once daily.
- CrCl <20 or severe hepatic: max 10 mg.

PHARMACOKINETICS

- **Onset:** 1–2 h. **Duration:** 24 h.
- Bioavailability ~90%; half metabolised in liver, half excreted unchanged in urine (balanced clearance).
- **t½:** 10–12 h.

PHARMACODYNAMICS

- Highly selective **β1 blockade** → ↓ HR, contractility, renin.
- In HF: protects heart from chronic sympathetic overstimulation → reverse remodelling.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fatigue, dizziness.
- Bradycardia, cold extremities.
- Nausea, headache.

ADVERSE EFFECTS (SERIOUS)

- Worsening heart failure during titration.
- AV block, severe bradycardia.
- Bronchospasm (less likely).
- Masked hypoglycaemia; withdrawal rebound.

CONTRAINDICATIONS & CAUTIONS

- Acute decompensated heart failure needing IV inotropes.
- Cardiogenic shock; 2nd/3rd-degree AV block; sick sinus syndrome; HR <60 before treatment.
- Severe asthma; severe peripheral arterial disease; untreated phaeochromocytoma.

KEY INTERACTIONS

- **Verapamil, diltiazem, amiodarone, digoxin:** bradycardia.
- **Clonidine:** rebound hypertension.
- **Antidiabetics:** masked hypoglycaemia.

NURSING MANAGEMENT

- Check HR and BP before giving; hold if HR <50–55 (per order).
- HF titration: daily weight, oedema, breathlessness.
- Report worsening HF symptoms.
- Never stop abruptly.
- Monitor glucose in diabetics.

PATIENT EDUCATION

- Symptoms may worsen briefly when dose increases; report weight gain or breathlessness.
- Do not stop suddenly.
- Take every morning.
- Rise slowly.

KEY POINTS

- Highly β1-selective; first-line in HFrEF.
- Check HR before dose.
- Start 1.25 mg; target 10 mg.
- Taper to stop.